

BONUS EXERCISE

```
In [ ]: from profilers import profile_cpu
```

To retrieve the CPU usage percentage per core for the diffusion codes

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In [ ]: @profile_cpu(interval=1)
def evolve(grid, dt, D=1.0):
    xmax, ymax = grid_shape
    new_grid = [[0.0] * ymax for x in range(xmax)]
    for i in range(xmax):
        for j in range(ymax):
            grid_xx = (
                grid[(i + 1) % xmax][j] + grid[(i - 1) % xmax][j] - 2.0 *
            )
            grid_yy = (
                grid[i][(j + 1) % ymax] + grid[i][(j - 1) % ymax] - 2.0 *
            )
            new_grid[i][j] = grid[i][j] + D * (grid_xx + grid_yy) * dt
    return new_grid
def run_experiment(num_iterations):
    # Setting up initial conditions
    xmax, ymax = grid_shape
    grid = [[0.0] * ymax for x in range(xmax)]

    # These initial conditions are simulating a drop of dye in the middle
    # simulated region
    block_low = int(grid_shape[0] * 0.4)
    block_high = int(grid_shape[0] * 0.5)
    for i in range(block_low, block_high):
        for j in range(block_low, block_high):
            grid[i][j] = 0.005

    # Evolve the initial conditions
    for i in range(num_iterations):
        grid = evolve(grid, 0.1)
```

The profile_cpu decorator wraps the evolve function

Every time evolve() is called, CPU usage is measured

What happens on each call to evolve

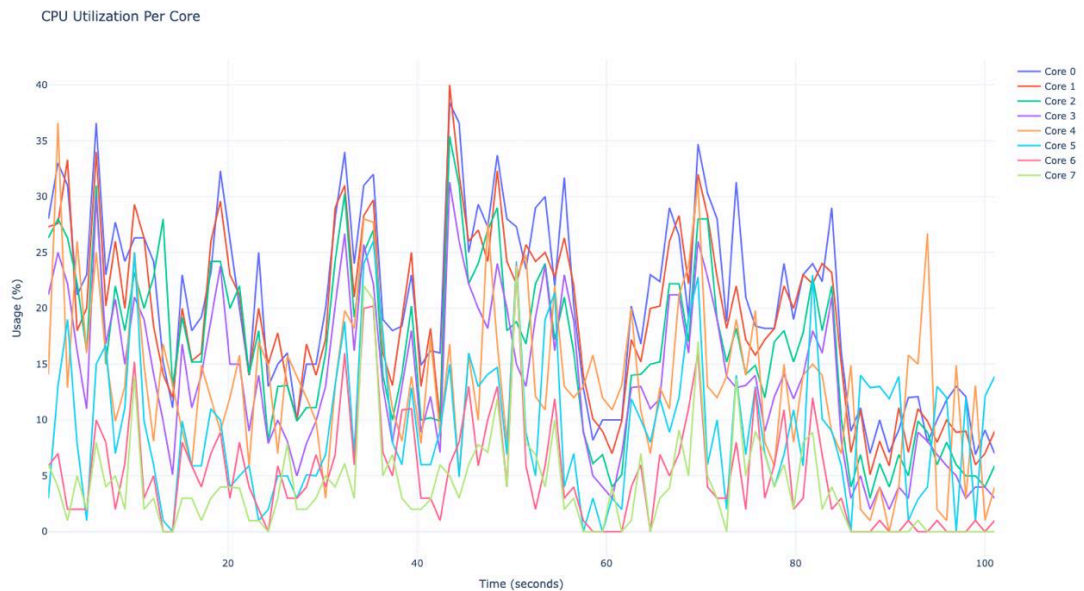
1. Before evolve starts
 - A global CUPProfiler is fetched (or created)
 - A background thread starts running
2. While evolve runs
 - every 1 second, it samples:

- CPU usage per core

- Current timestamp

3. After evolve finishes

- The profiler thread is stopped
- The collected samples are kept in memory



While the diffusion code is running, the CPU is working steadily but not very hard. All cores show some activity, but none of them are fully used

The table shows the summary of the CPU usage

Time (s)	Core 0	Core 1	Core 2	Core 3	Core 4	Core 5	Core 6	Core 7
1.01	28	27.3	26.3	21.2	14.1	3	5.9	6
2.01	33	27.6	28	25	36.6	13	7	4
3.03	31	33.3	26.3	22.2	12.9	19	2	1
4.04	21.2	18	22.2	16.2	26	7.9	2	5
5.05	23	20	16.2	11	16	1	2	2
6.06	36.6	34	31	30	25	15	10	8
7.07	23	20.2	15	16.8	15.8	16.7	8	4
8.08	27.7	26	22	20.8	9.9	7	2	5
9.09	24.2	20	18	15	13	11.1	6.1	2
10.1	26.3	29.3	23.2	21	25	25	15.2	13.9
11.11	26.3	26.3	20	19	14.9	9.9	3	2

Time (s)	Core 0	Core 1	Core 2	Core 3	Core 4	Core 5	Core 6	Core 7
12.12	24.2	18.4	22.8	14.1	8.1	6	5	3
13.13	15.2	14.1	28	9.9	16.8	1	0	0
14.14	11.1	12	13	5.1	13.1	0	0	0
15.15	23	20	19.2	16.8	9	9.9	8	3
16.16	18	15.3	15.2	11.1	5.9	5.9	5.9	3
17.17	19.2	16	15.2	14	14.9	5.9	4	1
18.18	23.2	26	24.2	19	12	11	7	3
19.19	32.3	29.6	24.2	23.8	9	10	8.9	4
20.2	26	23	20	15	12	4	3	4
21.21	20	21.2	22	15	15.8	5	8	3.9
22.22	14	14	14.1	9	5.9	5.9	4	1
23.23	25	20	18	14	17	1	2	1
24.24	13	15	8.1	7.9	14.9	2	0	0
25.25	15	17.8	13	10	7	5	5.9	3
26.26	16	13.1	13.1	8.1	15.8	5	3	7.9
27.27	10	10.1	9.9	5	13.9	3	3	2
28.28	15	16.8	11.1	7.9	12	5.1	4	2
29.29	15	14	11.1	9.9	9.9	5	6.9	3
30.3	20	17.2	16	13	3	6.9	4	5
31.31	28	29	24.2	20.2	13	13.1	6.9	4
32.33	34	31	30.3	26.7	19.8	18.8	16	6.1
33.33	24	21	19.2	16.2	18.2	7	6	3
34.34	31	28.3	25	25.7	28	24	20	22
35.35	32	29.7	27	22.2	27.7	26	20.2	20.8
36.36	19	16	14	13	16.8	17.8	7.1	5
37.37	18	13.1	10	8	11	8	5	6.9
38.38	18.4	19	14.1	13	8.1	6	10.9	3
39.38	23	25	20.2	18	13.9	12.9	11	2
40.39	14.9	13	10	8.9	7.9	6	3	2
41.4	16.2	18.2	10.2	12.1	17.3	6	3	2.9
42.41	16	10	9.9	7.1	8.8	9	1	6
43.42	38.4	40	35.4	31.3	16.8	15	6	5
44.42	36.6	32	31	26	5	4.9	8	3
45.43	25	26	22.2	22.2	15.8	16	13	6

Time (s)	Core 0	Core 1	Core 2	Core 3	Core 4	Core 5	Core 6	Core 7
46.44	29.3	27	24	20	10	13	5.9	7.8
47.45	27.3	24.2	27	18.2	27.7	14.1	10	7.1
48.46	33.7	32.3	29	24	16.8	14.7	13	11.9
49.47	28	24.2	18	20	8	6.9	4	4
50.48	27.3	22.2	18.8	15	22.8	24.2	24	23.8
51.49	23.5	25.7	16.8	13	24.8	8.9	5.9	7.9
52.5	29	24.2	22.2	19.2	12.1	5	2	6.9
53.52	30	25	24	23.8	10.9	19	5.9	4
54.53	22	22.8	17.2	16.2	22	21.4	11.9	10
55.54	31.7	26.3	21	23	13	4	3	2
56.55	20.2	22	16	18.8	12	7	4	3
57.56	12.9	14	8.9	9	13	0	1	0
58.57	8.2	10.1	6.1	5	15.8	3	0	0
59.58	10	9	6.9	4	12	0	0	0
60.6	10	7	4	3	10.9	3	0	4
61.61	10	9.9	5.1	7	13.1	2	0	0
62.62	20.2	17.2	14	12.9	20	11.9	4	1
63.63	16.8	15.2	14.1	13	10.9	10	6	7
64.64	23	20	15	11	7	8	0	0
65.65	22.4	20.2	15.2	11.9	12.9	11.9	6.9	3
66.66	29	26	22.2	21.2	11	8.9	5	3.9
67.66	26.5	28.3	22.2	21.2	21.2	12	7	9.1
68.67	19.2	22.2	17	16	23.8	19.2	11	5
69.68	34.7	32	28	26	31.3	22.8	16	17
70.69	30.3	28.3	28	22.8	13	6	4	5
71.7	28	22.8	19	19	12	10	3	3
72.71	18.8	18.2	15.2	13.9	13.9	2	3	0
73.72	31.3	22	18.2	12.9	19	14	8	13.7
74.73	21	17.2	14.1	13.1	14	6.9	2	5
75.74	18.4	15.8	14.9	14	19.8	13	12.9	9
76.75	18.2	17.2	12	9	8	7	3	6.9
77.76	18.2	18.2	17	12.1	5.9	4	6	4
78.77	24	22	18	14.1	15	6.9	10.9	6
79.77	19	20	15.2	11.9	8	10.9	2	2

Time (s)	Core 0	Core 1	Core 2	Core 3	Core 4	Core 5	Core 6	Core 7
80.78	23	23	17.8	14	14	5.9	3	8
81.8	24	22.2	22.8	18	15	23	12	8.9
82.81	22.4	24	18	16	14	10.1	6.9	2
83.82	29	23.2	22	21	8.9	9	2	4
84.83	16	15	10.1	8.1	7	5.9	3	2
85.84	9	7.1	4	3	14.9	0	0	0
86.85	11.1	10.9	6.9	5	2	14	0	0
87.87	7	5.1	3	2	1	12.9	0	0
88.88	10	8.1	6.1	4	4	13	1	0
89.89	7.1	5.9	4	2	0	11.9	0	0
90.91	9.1	11.1	6.9	4	4	13.9	0	0
91.92	12	7.1	5	3	15.8	1	1	0
92.93	12.1	11	9.9	8.9	15	2.9	0	1
93.94	8	9.9	8.9	8.1	26.7	4	0	0
94.95	10	8	6	6.9	2	13	1	0
95.96	11.8	10	8	5.9	1	12	0	0
96.98	13	8.9	6	5	14.9	0	0	0
97.99	12.1	9	5	3	3	12	0	0
99	6.9	6	5	4	13.1	1	1	0
100.02	9.1	7	4	4	1	12.1	0	0
101.03	7	9	5.9	3	4	13.9	1	0

The table shows that the diffusion simulation keeps the CPU moderately busy over time, with work spread unevenly across the cores. A few cores consistently do more work, while others remain lightly used or idle. This indicates that the computation is heavy but largely runs in a single-threaded manner, with limited parallel execution.

Note: The Jupyter notebook was unable to run successfully due to certain dependencies that could not be migrated to the notebook environment. As a result, the code execution fails within Jupyter. For the complete and working implementation, please refer to the code provided in the accompanying ZIP file.

In []: